Chest X-ray Pneumonia Detection System

Team 19 - CSAI

An AI-Assisted Web Platform for Pneumonia Screening and Localization from Chest Radiographs

Supervisor: Prof. Dr. Khaled Mostafa

Moamen Elsayed Elsharkawy (202202015) . Habiba Ayman Amin (202202088) . Ahmed Gamal Abdelfattah (202201638) . Sara Mostafa Ali (202201305)

School of Computational Sciences and Artificial Intelligence (CSAI), Zewail City of Science and Technology - June 2026

Problem and Motivation

Pneumonia is a leading cause of illness and death worldwide, and the chest X-ray is the first-line tool to detect it. In many clinics the number of radiographs far exceeds the radiologists available to read them, which delays diagnosis and raises the risk of missed cases. A missed pneumonia is the most dangerous error. We provide a fast, accessible AI second reader that flags and localizes pneumonia while the doctor stays in control.

Objectives

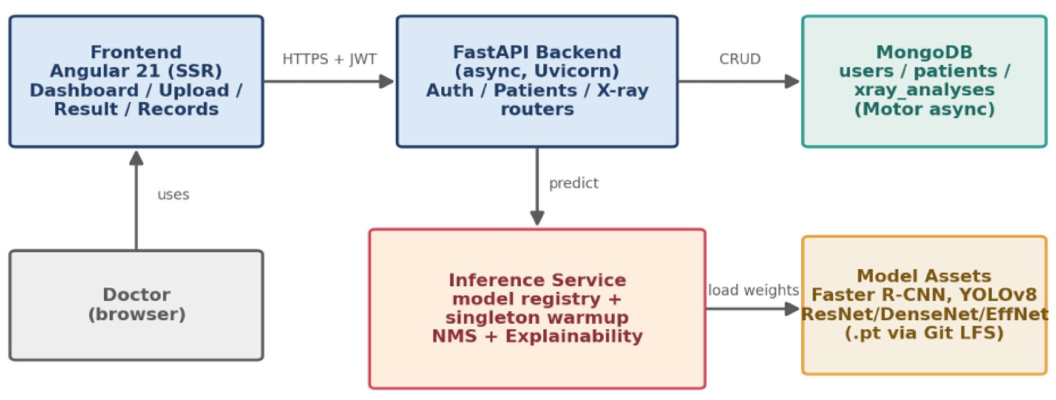
- Train and honestly evaluate models for pneumonia detection and localization.
- Deliver a secure, multi-user web app with patient management and history.
- Provide visual explainability with every prediction.
- Keep the system reproducible, documented, and maintainable.

Proposed Solution

A doctor signs in, manages patients, uploads a chest X-ray, and receives an automated reading: a pneumonia decision, a bounding box, a confidence score, and an explainability heatmap. Five models are served behind one registry; Faster R-CNN is the deployed default. Every analysis is stored per patient, and the physician confirms every result.

System Architecture

System Architecture



Technologies Used

Frontend: Angular 21 (SSR). Backend: FastAPI (async), Uvicorn.
Database: MongoDB (Motor). AI/ML: PyTorch, torchvision, Ultralytics.
Security: JWT, bcrypt, CORS. Model storage: Git LFS.

Key Numbers

0.886
Best AUC (EfficientNet-B0)

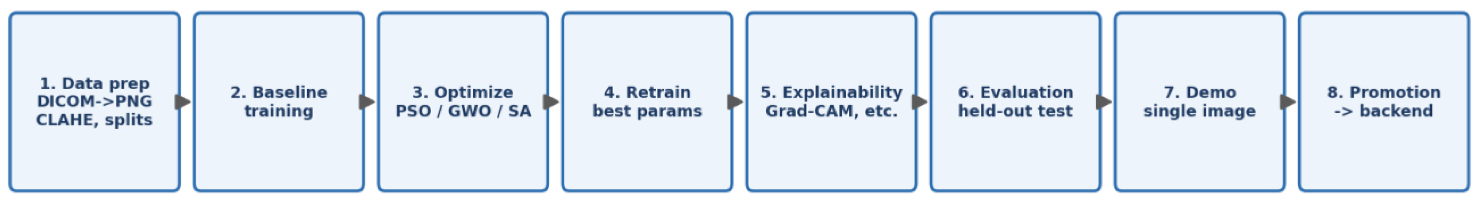
0.812
Recall (Faster R-CNN)

5
Deep-learning models served

26,684
RSNA radiographs

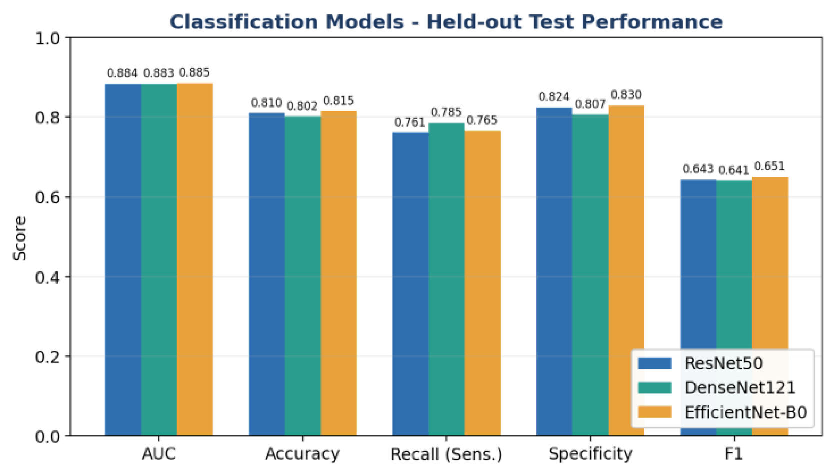
Methodology

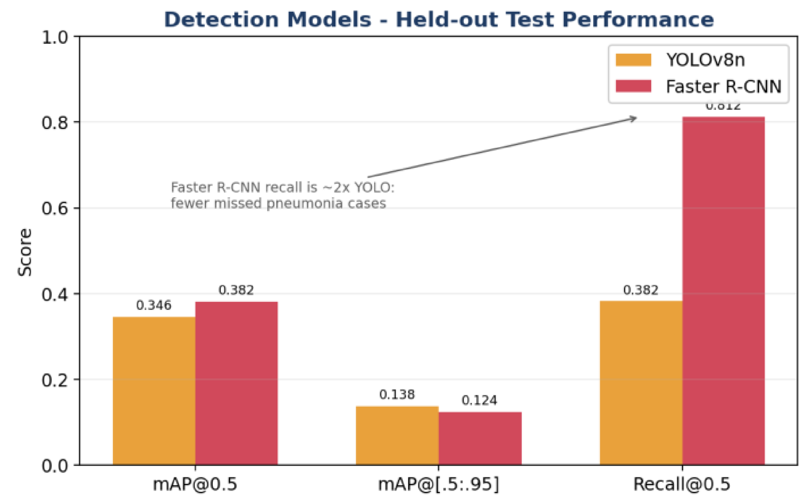
AI / ML Pipeline (8 phases)



RSNA Pneumonia Detection Challenge (~26,684 radiographs), DICOM to PNG, leak-free patient-wise split (20% held-out test: 4,135 normal / 1,202 pneumonia). Transfer learning, mixed-precision training, augmentation, and early stopping; PSO/GWO/SA hyperparameter search.

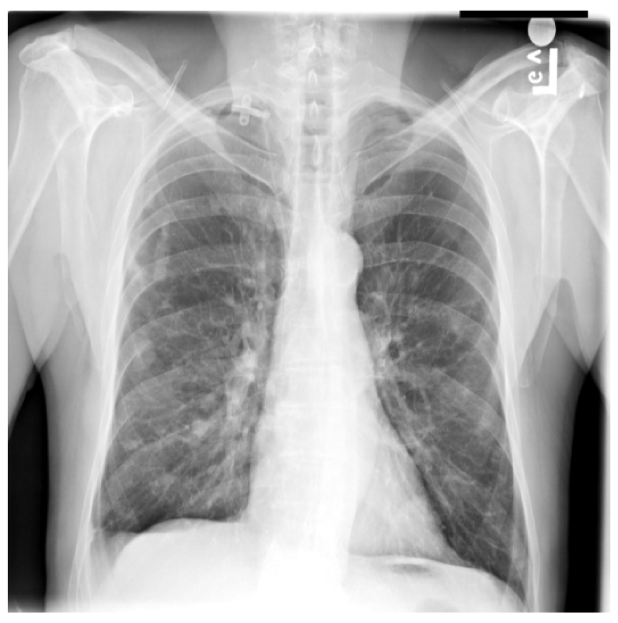
Results (held-out test set)





Best classifier EfficientNet-B0: AUC 0.886. Deployed detector Faster R-CNN: recall 0.812 (mAP@0.5 0.381). Recall is prioritized because a missed pneumonia is the costly error. Numbers are honest and competitive with the literature (RSNA detection mAP typically 0.32-0.39).

Sample Output and Explainability



Every prediction is accompanied by a bounding box and a heatmap (Grad-CAM and related methods) so the clinician can verify that the model is looking at the lungs, not an artifact.

Conclusion

A complete, working, explainable full-stack pneumonia screening system: five models rigorously evaluated, secure and reproducible, demonstrating the full path from research model to usable clinical product.

Project Repository

github.com/202201638/Graduation_project_Fully_team_2026